Bengali ASR: six models on FLEURS

920 FLEURS bn_in TEST utterances · greedy, no external LM · batch 1 on one RTX A5000 · ‡ batched with vLLM on an A100

Model	WER	95% CI (clustered)	CER	ms/clip	Params
Conformer Large (hishab)	14.37%	[13.11–15.73]	4.48%	77	121.5M
Voxtral Mini bn-BD ‡	14.73%	[13.61–15.92]	4.77%	p. 5	4,676M
Whisper Medium	16.23%	[15.05–17.45]	5.09%	1,866	763.9M
▸ **ehzawad FastConformer-CTC**	**17.12%**	**[15.95–18.34]**	**5.12%**	**83**	**115.6M**
Wav2Vec2	20.64%	[19.46–21.89]	6.02%	64	315.5M
hishab FastConformer	36.22%	[33.88–38.72]	9.05%	79	115.6M

Test split only. The ehzawad model trained on a corpus that includes the FLEURS train split, and 88 of the 150 validation-split sentences also occur there; the 402 validation recordings are therefore excluded for every model and the table is the 920-utterance test split.

Reading this table

ehzawad/stt_bn_fastconformer_ctc is a FastConformer-CTC large (115.6M parameters, 8x subsampling, 1,024-piece Bengali BPE) trained on 1,692 hours of human-supervised Bengali, CC-BY-SA-4.0. The other four models are public checkpoints from hishab and SayedShaun (mirrors of tugstugi's Whisper and qdv206's wav2vec2 competition models).

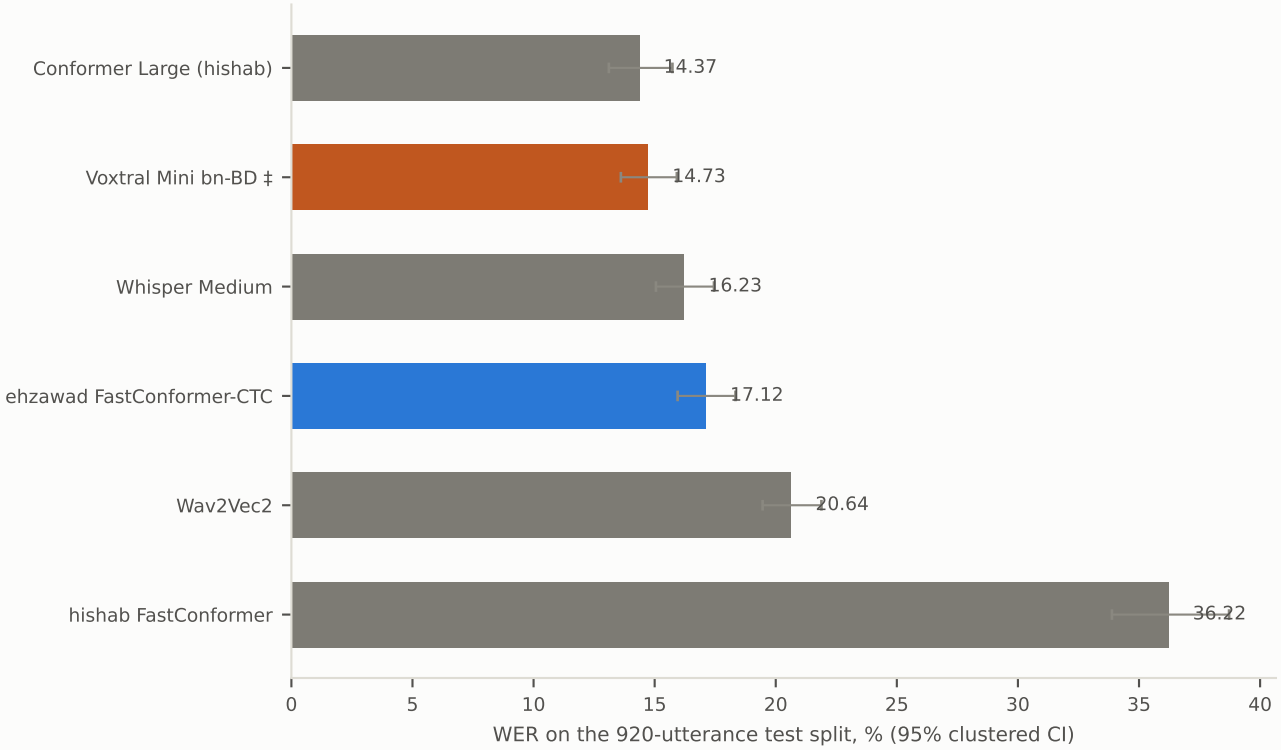
Paired sentence-clustered deltas for the highlighted model (10,000 draws):

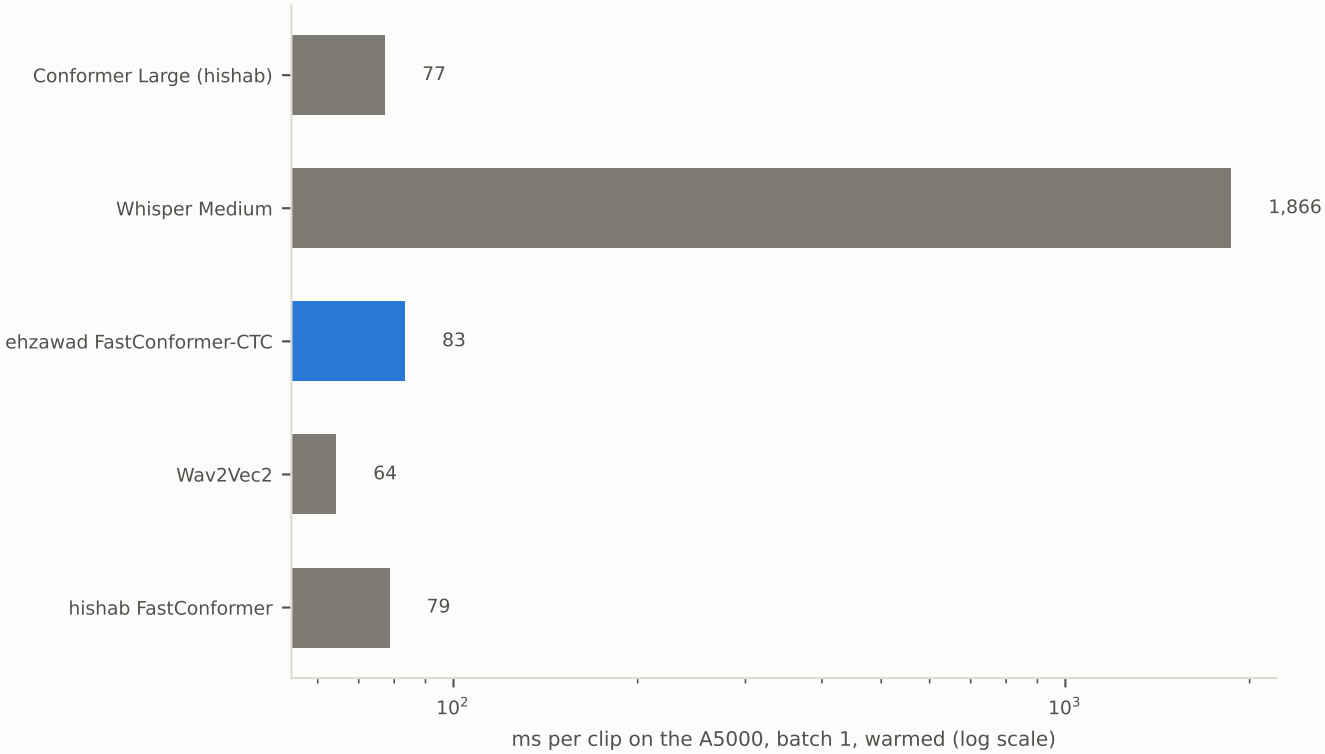
vs Whisper Medium +0.89 pp [+0.27, +1.50]
vs Conformer Large +2.74 pp [+2.01, +3.46]

Under a point behind Whisper Medium at 22x its speed; 2.7 points behind Conformer Large. The highlighted model had also been evaluated on FLEURS test during its own development, so read its FLEURS row as a like-for-like re-measurement and the other corpora as the fresh evidence.

‡ Voxtral Mini bn-BD is a private fine-tune of Voxtral Mini 3B whose card documents no training data. It went through the same asr_core path and scorer but was decoded with vLLM in continuous batches on an A100-40GB, a different stack from the rows above; its batch-1 latency is on page 5. Against Conformer Large on this split: +0.35 pp [-0.36, +1.06], inconclusive.

FLEURS test: accuracy and speed side by side





The two orderings are not the same ordering. The three NeMo CTC models sit within 10% of each other in speed; Whisper Medium costs ~22x per clip at batch 1 for its accuracy. Voxtral Mini was not timed on the A5000; page 5 times all six on one A100.

Beyond FLEURS: five more official test sets

Frozen seeded 1,000-utterance subsets of the OFFICIAL test splits · one model at a time · same backends and scorer

Set	official rows	sampled	audio	ref. words	stratified by
Vaani test	12,695	1,000	1.26 h	10,082	29 state/district strata
SPRING-INX R1	1,879	1,000	2.54 h	19,706	Latin / no-Latin reference
SPRING-INX R2	1,872	1,000	2.61 h	20,353	Latin / no-Latin reference
Kathbath known	2,806	1,000	1.79 h	10,801	speaker gender
Kathbath unknown	1,783	1,000	1.81 h	10,887	speaker gender
IndicVoices conv.	3,603	1,000	1.57 h	12,918	scenario (conversation / extempore)

Model (WER %)	Vaani	SPR R1	SPR R2	Kb known	Kb unk	IV conv	flags
▸ **ehzawad FastConformer-CTC**	**24.66**	**39.24**	**38.41**	**13.38**	**13.43**	**12.61**	**N N N S S S**
Whisper Medium	26.25	39.96	39.70	9.69	9.79	25.03	U U U U U U
Conformer Large (hishab)	33.10	42.46	42.24	9.33	8.79	29.68	N N N S S N
hishab FastConformer	28.71	46.47	45.08	25.81	29.87	30.14	U U U U U U
Wav2Vec2	31.26	47.28	47.33	19.08	18.80	33.01	U U U U U U
Voxtral Mini bn-BD ‡	22.12	36.99	40.04	10.73	10.89	13.45	U U U U U U

flags, per set in the column order: N = corpus not in the model's declared training data; S = trained on a sibling split of the same corpus (Kathbath train; Conformer Large also its validation split); U = training provenance not documented. None of these means "verified clean". The highlighted model's Vaani and SPRING hypotheses come from its earlier evaluation of the full official splits (same greedy decoding, batched), scored here on exactly these utterances; they are not re-timed. Its Kathbath and IndicVoices rows are batch-1 decodes here. IndicVoices conversational is NOT a clean head-to-head for it: it trained on the IndicVoices train split (same collection, prompts and transcription conventions, different speakers), which is a large part of its 12.6% there.

What the table says

On the two corpora nobody trained on, the FLEURS ranking does not hold. Conformer Large - the FLEURS leader at 14.4% - is last on Vaani (33.1%) and third on both SPRING sets; its model card lists read and broadcast corpora and neither Vaani nor SPRING-INX. ehzawad/stt_bn_fastconformer_ctc leads the five public models on Vaani and both SPRING sets; Whisper Medium is second on all three at 13-20x the latency of the NeMo CTC models. The private Voxtral Mini fine-tune (‡, U on every set) is lower still on Vaani and SPRING R1 (page 5).

On Kathbath the order flips back: Conformer Large (9.3 / 8.8%) and Whisper (9.7 / 9.8%) lead and the highlighted model is third (13.4 / 13.4%). Conformer Large trained on Kathbath validation as well as train, Whisper's provenance for Kathbath is unknown (a U cell can hide training overlap), and the highlighted model saw only the train split. Known- vs unknown-speaker Kathbath differ by under a point for every model except hishab FastConformer. On IndicVoices conversational the four models that never saw the collection read 25-33%, Conformer Large (the one N cell) 29.7% with 7 empty hypotheses, hishab FastConformer 21 and Wav2Vec2 33 empties. Sentence-clustered 95% intervals for every cell are in the README and summary.

Latency per set, and how the runs were controlled

SPRING note: about half of its reference utterances contain English words in Latin script; none of the six models emitted a single Latin-script word, so those words are errors for every model alike. On the Latin-free half the highlighted model (35.4 / 35.1%) and Whisper Medium (35.9 / 36.7%) are within each other's intervals (per-model split in outputs_multiset/summary_multiset.json).

ms per clip, batch 1, warmed, default runtime (N/A = stored hypotheses or batched decodes, not timed here)

Model	Vaani	SPR R1	SPR R2	Kb known	Kb unk	IV conv
▸ **ehzawad FastConformer-CTC**	**N/A**	**N/A**	**N/A**	**72**	**70**	**74**
Whisper Medium	919	1,454	1,494	1,014	1,041	1,034
Conformer Large (hishab)	74	75	76	70	71	75
hishab FastConformer	72	74	73	70	71	73
Wav2Vec2	33	49	51	35	35	39
Voxtral Mini bn-BD ‡	N/A	N/A	N/A	N/A	N/A	N/A

Protocol. Every set is a frozen subset (seed 20260915; proportional stratification; per-file SHA-256 verified before decoding). One model resident at a time on the A5000; no other process used the card during any run. Manifests: multiset/; raw predictions and per-run GPU state: outputs_multiset/; harness bench_run_set.py (bench_run.py's backends unchanged); scorer score_multiset.py. Reproduce the scoring from the committed predictions. Voxtral Mini was decoded on an A100-40GB with vLLM (bench_run_voxtral_vllm.py), same manifests and SHA-256 checks.

Evaluation data: FLEURS (Google, CC-BY-4.0); Vaani transcription part (ARTPARK-IISc, CC-BY-4.0); SPRING-INX R1/R2 (SPRING Lab, IIT Madras, CC-BY-4.0); Kathbath (AI4Bharat IndicSUPERB, CC0); IndicVoices (AI4Bharat, CC-BY-4.0). Models: hishab (Conformer Large, FastConformer); SayedShaun (Whisper Medium mirror of tugstugi's competition model; wav2vec2 mirror of qdv206's); ehzawad (FastConformer-CTC; Voxtral Mini bn-BD, private, fine-tuned from mistralai/Voxtral-Mini-3B-2507, Apache-2.0). Author of this benchmark: Emrul Zawad.

One A100: latency of all six, and Voxtral Mini bn-BD

920 FLEURS test clips · batch 1 · warm-up discarded · each clip timed between CUDA syncs, audio loading and staging included

Model	mean ms	median	p95	x real time	same transcripts
Wav2Vec2	56	54	86	242x	913/920
Conformer Large (hishab)	87	84	97	155x	920/920
ehzawad FastConformer-CTC	88	85	99	152x	919/920
hishab FastConformer	89	85	100	151x	919/920
Voxtral Mini bn-BD ‡	438	421	681	31x	860/920*
Whisper Medium	1,945	1,837	3,135	7x	916/920

These time the configured pipelines, not the models in isolation: NeMo and transformers (Whisper in FP16) from bench_run.py's backends at the pinned revisions, torch 2.11; Voxtral Mini through vLLM 0.31 with CUDA graphs, torch 2.13. One pass each; a few milliseconds apart is not an established ordering. A low-priority scoring process shared the CPUs during the four fastest runs (effect not measured). "same transcripts" compares each model's A100 output with its A5000 predictions; * Voxtral Mini's with its own batched run.

Voxtral Mini bn-BD minus the best public model per set

Set	best public model	WER points	95% CI	paired interval
FLEURS test	Conformer Large (hishab)	+0.35	[-0.36, +1.06]	inconclusive
Vaani test	ehzawad FastConformer-CTC	-2.54	[-3.20, -1.88]	lower
SPRING-INX R1	ehzawad FastConformer-CTC	-2.25	[-2.86, -1.65]	lower
SPRING-INX R2	ehzawad FastConformer-CTC	+1.63	[-1.49, +5.78]	inconclusive
Kathbath known	Conformer Large (hishab)	+1.40	[+0.74, +2.08]	higher
Kathbath unknown	Conformer Large (hishab)	+2.11	[+1.43, +2.80]	higher
IndicVoices conv.	ehzawad FastConformer-CTC	+0.84	[-0.01, +1.60]	inconclusive

Voxtral Mini bn-BD (4.7B parameters) is a private fine-tune of mistralai/Voxtral-Mini-3B-2507; its card documents no training data, so every cell is U. It used the benchmark's manifests, SHA-256 checks, asr_core path and scorer, but its accuracy run was decoded with vLLM 0.31 in continuous batches (eager, up to 256 pieces) on an A100-40GB, unlike the batch-1 A5000 decodes of the public models. It writes numbers as Bengali words, which the scorer counts as errors against digit references, as it does for every model. On SPRING-INX R2, 3 recordings ran to the 500-token cap repeating a word; they contribute 663 of its 8,150 word errors and stay in its 40.04% (without them 36.86% on 997 recordings, a conditional diagnostic, not a corrected score). Its batch-1 timing run (CUDA graphs) gave the same transcript as its batched run on 860/920 clips, with the same 2,616 word errors: outputs depend on the engine configuration. Intervals are pointwise, not adjusted for multiple comparisons; an inconclusive result is not evidence of equivalence.